

Project Design Phase-II

Technology Stack / Architecture Stack

Date	04-11-2025
Team ID	13CF53BB1BAAF3193364A4F11DD1A083
Project Name	Lease Management System Using Salesforce

Technical Architecture Overview

The architecture for the Lease Management System leverages Salesforce’s cloud-based CRM platform with integrated declarative and programmatic automation, robust data modeling, and secure, scalable infrastructure.

Salesforce Lease Management:
Technical Architecture Overview





The diagram illustrates the Salesforce Lease Management Technical Architecture. At the center is the 'Salesforce Cloud Platform Database'. It is connected to several components: 'User Interface' (Lightning Experience), 'Application Logic' (Custom Objects & Validation, Lightning Formdate), 'Communication Layer' (Outbound, Email Templates, Apex Triggers, Scheduled Jobs), 'File Storage', 'Analytics Tools', and another 'Communication Layer' (Outbound, Payment Gateways). Arrows indicate the flow of data and interactions between these components and the central database.

Table 1: Components & Technologies

User Interface	Description	Technology Usd
User Interface	Guston	5%
	Constinct	
Custom Objects & Validation	Fumistitre	1,320%
Flows & Approval Proceses	Homisttre	20%
Apex Triger ed Jobs	Conndurled Jobs	1,530%
Cne Vestore Jobs	Treemnicst	
Scaaant Architecture	Irlangal ture	1,5,50%
Performance	Altolytics	1,30%
Infrastructure Server	Conteint	

Table 2: Application Characteristics

Security	Description	Technology Usd
Security	Description	20%
Scalable Architecture	Gecurity	650%
Payment Architecture	Titeh	500%
Payment Gatevas	Nouiatel	350%
	Technology	40%
Scalable Architecture	Titen	2,00%
Performance	Nourtk	1,20%
Performance	Gnanal	10%

Additional Notes

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Table 1: Components & Technologies

S.No	Component	Description	Technology Used
1	User Interface	Admins and users interact via Salesforce Lightning Web UI	Salesforce Lightning Experience, App Tabs
2	Application Logic-1	Data model, validation, field/update and assignment management	Salesforce Custom Objects, Fields, Validation Rules
3	Application Logic-2	Process automation, approval and leave workflows	Salesforce Flows, Approval Processes
4	Application Logic-3	Automated business logic for data integrity	Apex Triggers, Scheduled Apex, Process Builder
5	Communication Layer	Automated notification and emails for payment, approval and reminders	Salesforce Email Templates, Flow Email Action
6	Database	Stores tenant, property, lease, payment, user, and audit data	Salesforce Cloud Platform Database
7	File Storage	Attachments (lease docs, payment receipts)	Salesforce Files
8	External API-1	Integration with payment gateways (if needed)	Salesforce REST API, External Services
9	Integration Layer	Future expansion to property portals or analytics tools	Salesforce Integration APIs, Outbound Messaging
10	Infrastructure Server	SaaS cloud infrastructure, managed by Salesforce	Salesforce (multi-tenant SaaS)

Table 2: Application Characteristics

S.No	Characteristic	Description	Technology
1	Open-Source Frameworks	Not applicable; platform is proprietary	Salesforce
2	Security	Role-based access, field-level security, audit trails	Profiles, Permission Sets, Sharing Rules
3	Scalable Architecture	SaaS-based, multi-tenant, scalable to thousands of users/records	Salesforce Cloud Platform
4	Availability	High availability, disaster recovery provided by Salesforce	Multi-region cloud, automated backups
5	Performance	Optimized via indexed objects, asynchronous flows, caching	Indexed Objects, Salesforce Flow, Apex Batch
6	Monitoring & Logging	Audit logs for all critical changes and approvals	Salesforce Setup Audit Trails

Additional Notes

- The technology stack is fully cloud-based, minimizing infrastructure management and maximizing reliability and security.
- Integrations and automation are designed to ensure easy extensibility and future-proofing for new business requirements (property types, payment integrations, document management).